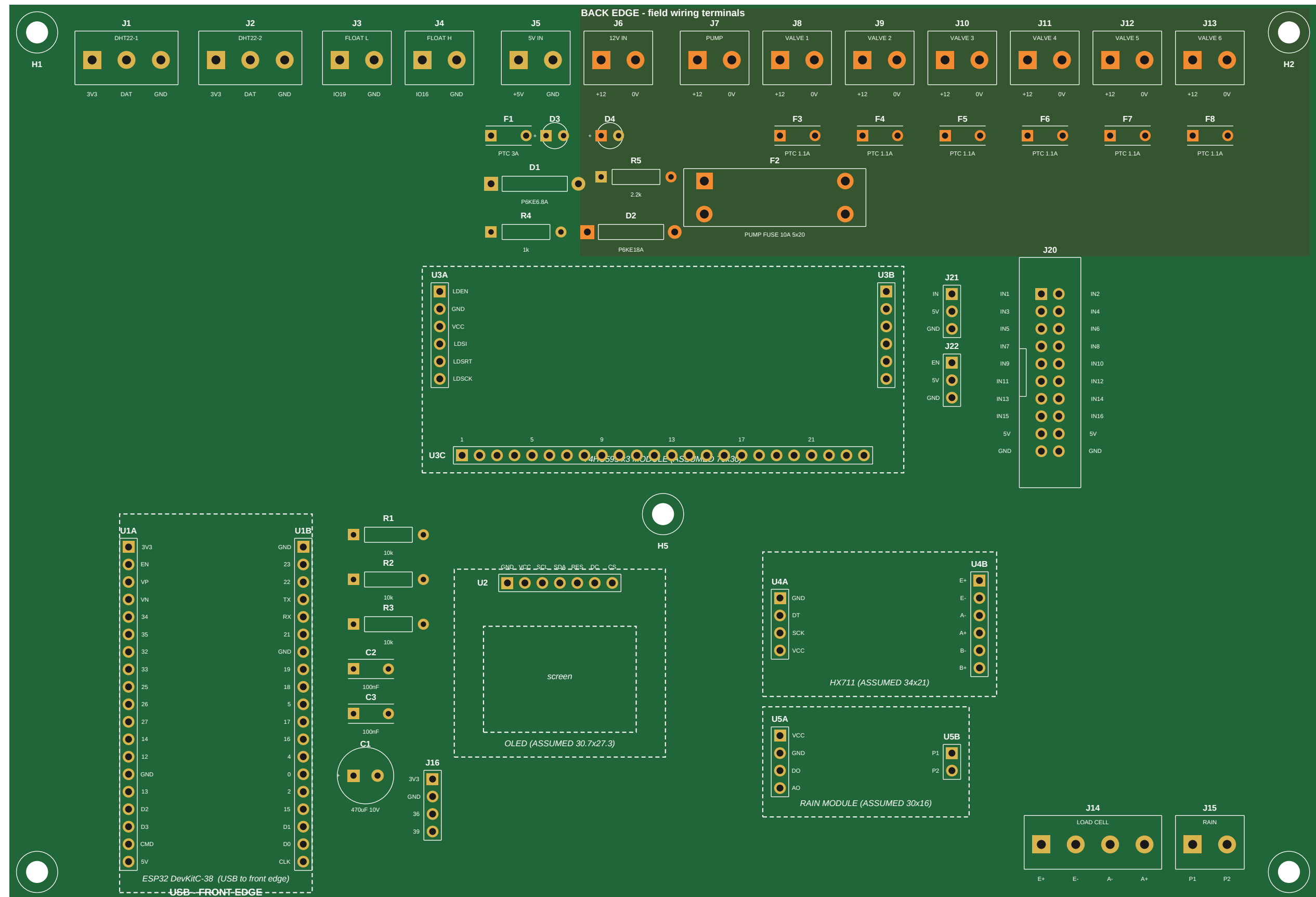


Room controller main PCB v2 - 190 x 130 mm, 2-layer, component side (faces the lid) - drawn 1.75 : 1



ASSUMED module sizes (replace with measurements):

- ESP32 DevKitC-38 (ESP32-32U): 55 x 28 mm, pin rows 25.4 mm apart
- DM0054 0.95in OLED (SSD1331): 30.7 x 27.3 mm, 7-pin row 2 mm in from the top edge, screen centre 14 mm below the pins
- 74HC595 x3 module (LD* pins): 70 x 30 mm: 6-pin input row on the left edge, 6-pin cascade row on the right edge, 24 outputs in one row along the front edge
- HX711 module (MD0020): 34 x 21 mm: 4-pin GND/DT/SCK/VCC row and 6-pin E+/E-/A+/B-/B+ row, 29 mm apart
- Rain comparator (MD0019): 30 x 16 mm: 4-pin VCC/GND/DO/AO row and 2-pin plate row, 25 mm apart
- MD0293 16-ch relay board: 180 x 90 x 20 mm, 5V, low-level trigger; inputs via 2x10 ribbon (IN1-16, 5V, GND)

Relay ribbon J20 (2x10):

pins 1-16 = IN1-IN16 (74HC595 Q0-Q15), 17-18 = +5V (relay VCC & JD-VCC), 19-20 = GND. Low level = relay ON. Relay 17 (Q16): J21 IN/5V/GND.

Bill of materials

- 2 x U1A,U1B: Female header 1x19 2.54mm, 8.5mm tall (ESP32)
- 1 x U2: Female header 1x7 2.54mm (OLED)
- 2 x U3A,U3B: Female header 1x6 2.54mm (595 in/cascade)
- 1 x U3C: Female header 1x24 2.54mm (595 outputs)
- 2 x U4A,U5A: Female header 1x4 2.54mm
- 1 x U4B: Female header 1x6 2.54mm (HX711 bridge)
- 1 x U5B: Female header 1x2 2.54mm (rain plate)
- 2 x J1,J2: Screw terminal KF301-5.0-3P
- 12 x J3-J13,J15: Screw terminal KF301-5.0-2P (17A)

- [illegible]